



Were the Haicheng and Tangshan Earthquakes in China in the 1970s Dragon-King Earthquakes?

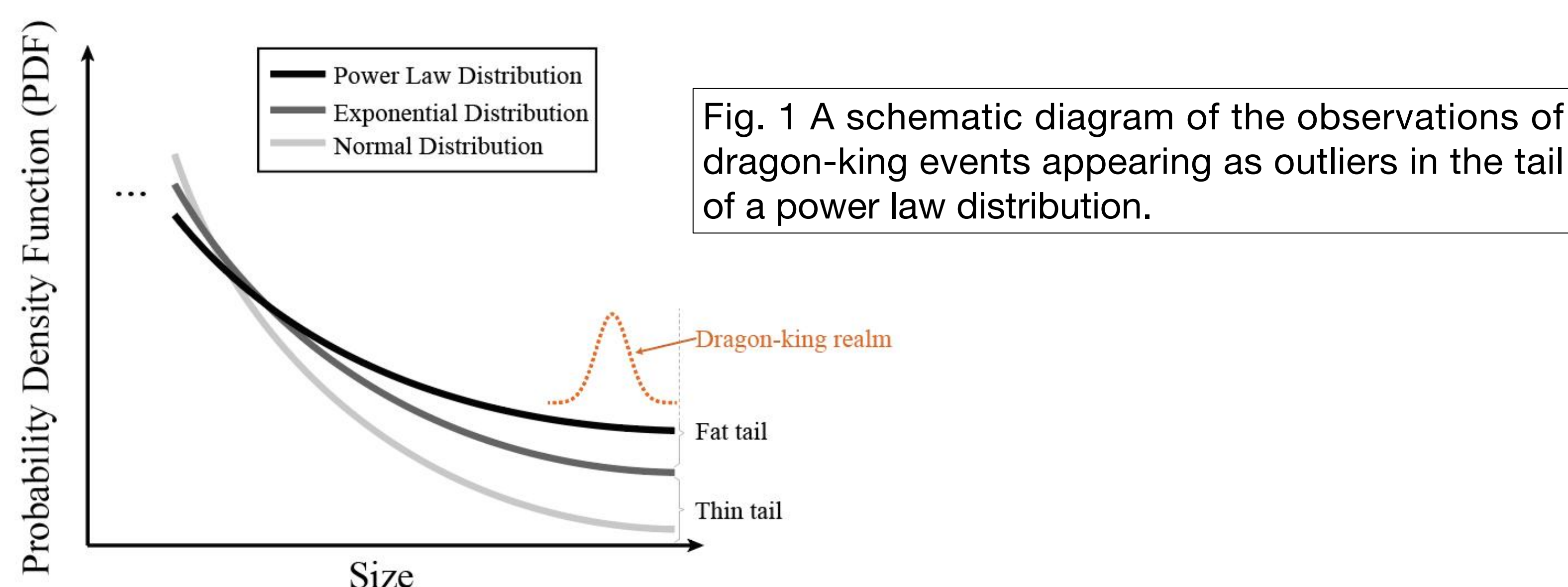
Jiawei Li, Didier Sornette

Institute of Risk Analysis, Prediction and Management (Risks-X), Southern University of Science and Technology (SUSTech)



Dragon-king events and dragon-king theory

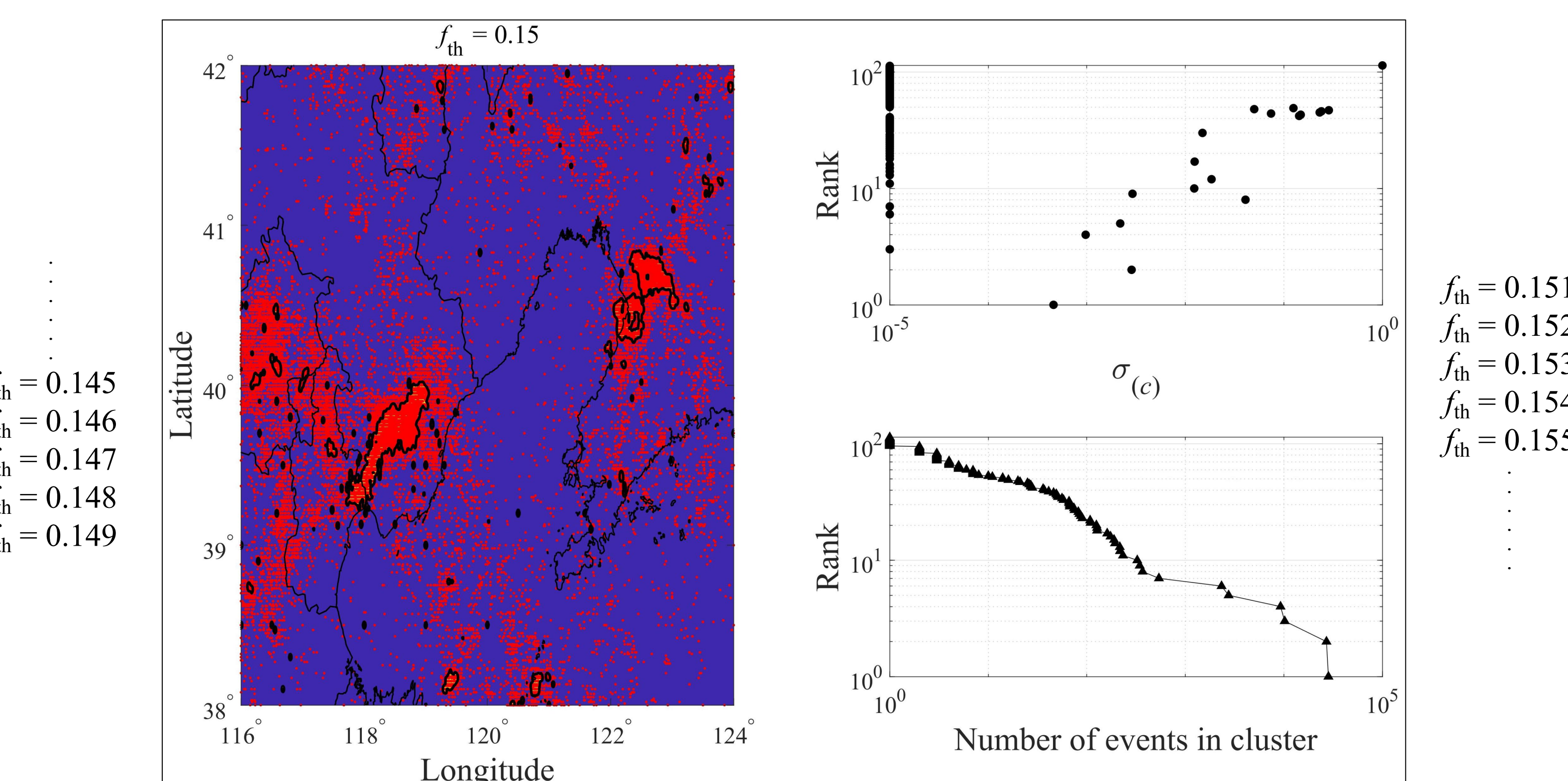
Dragon-kings (DK) embody a dual metaphor, expressing that an event is both exceptionally large ("king") and originates from unique circumstances ("dragon") compared to its peers. The hypothesis proposed by Sornette (2009), with a premonitory discussion emphasizing the predictability of catastrophes that are outliers (Sornette, 1999), posits that DK events arise from distinct mechanisms that occasionally amplify extreme events, resulting in the creation of runaway disasters on the downside as well as extraordinary opportunities on the upside (de S. Cavalcante et al., 2013; Sornette, 2009; Sornette and Ouillon, 2012).



Framework for detecting dragon-king earthquake

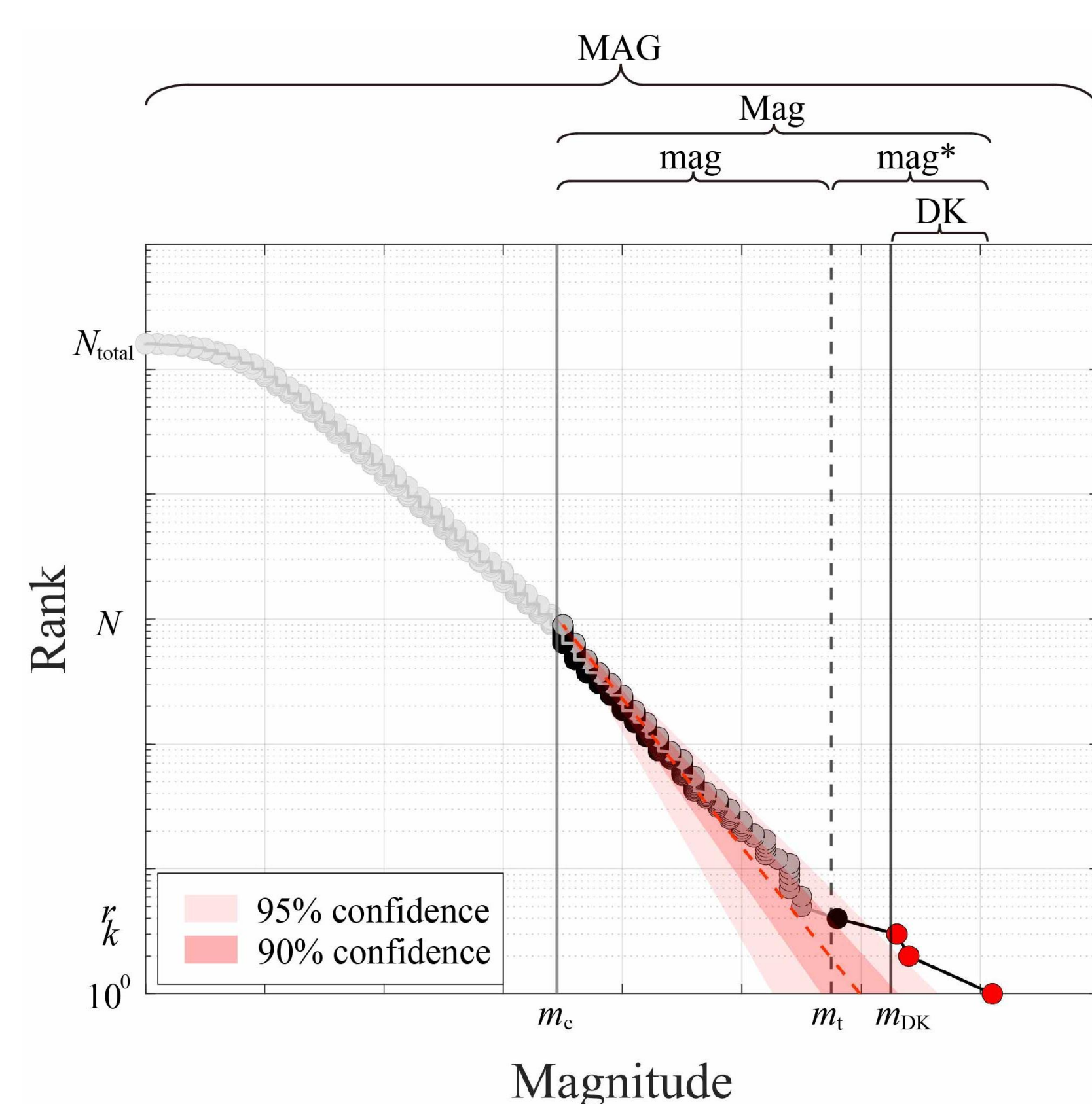
1. Defining the Study Region

$$f_{[T_1, T_2]}(\text{Lon}, \text{Lat}) = \frac{1}{S_{\text{norm}}} \sum_{t_i \in [T_1, T_2]}^{N_{\text{raw}}} \frac{1}{2\pi h_i^2} \exp \left[\alpha m_i - \frac{(\text{Lon} - \text{Lon}_i^{\text{eq}})^2 + (\text{Lat} - \text{Lat}_i^{\text{eq}})^2}{2h_i^2} \right]$$



2. Identifying Dragon-King Candidates

$$D^2(m_t; \text{Mag}) = -N - 2 \sum_{i=1}^N \left\{ \frac{2i}{2N+1} \log_{10} \left[\frac{\text{rank}(m_{(i)})}{N} \right] + \left(1 - \frac{2i}{2N+1} \right) \log_{10} \left[\frac{1 - \text{rank}(m_{(i)})}{N} \right] \right\}$$



3. Rigorous Statistical Testing

$$T_{j, N-v}^{\text{MRS}} = \frac{m_{(j)}}{\sum_{i=1}^{N-v} m_{(i)}}$$

4. Validating dragon-king earthquakes

$$T_{k, N-v}^{\text{SRS}} = \frac{\sum_{i=N-k+1}^N m_{(i)}}{\sum_{i=1}^{N-v} m_{(i)}}$$

Motivation

Whether large earthquakes enjoy some distinguishable properties that could make them predictable?

Are Haicheng and Tangshan Earthquakes Dragon-Kings?

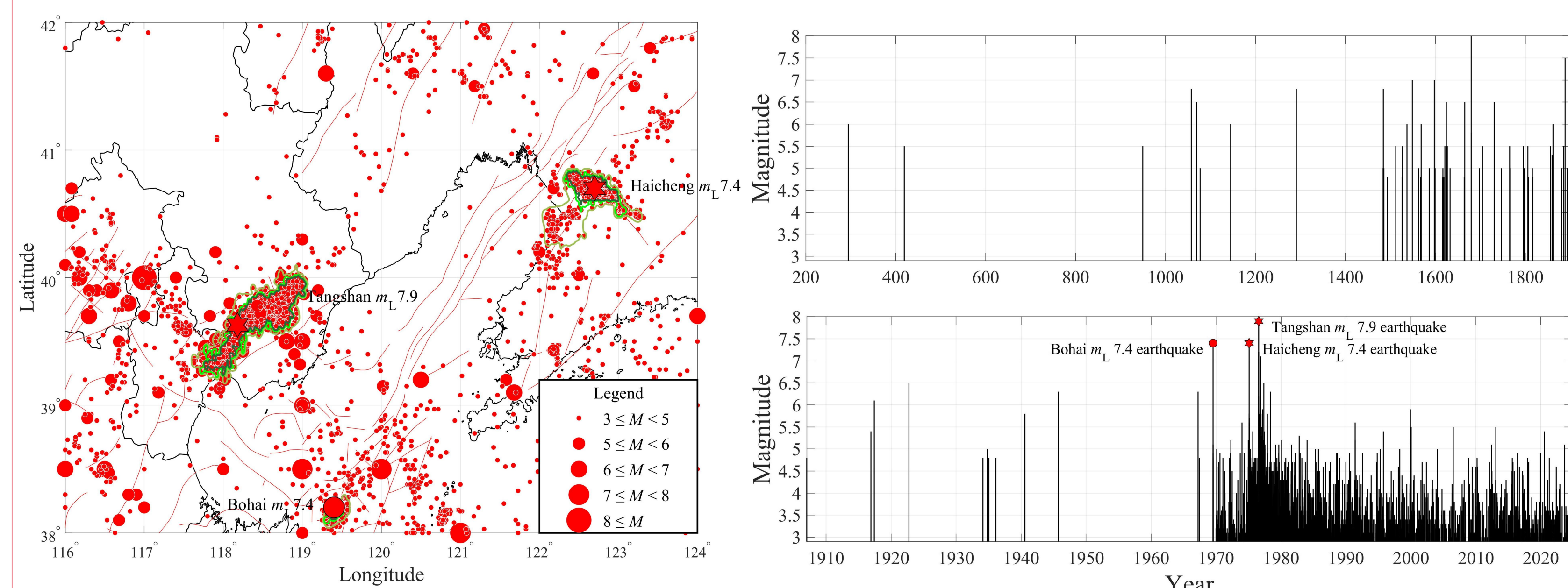


Fig. 2 Epicenters of earthquakes with $m_L \geq 3.0$ that occurred from 200 to 2024 in the vicinity of Bohai Bay, China from the China Earthquake Networks Center (CENC). Thin black lines indicate the provincial administrative boundaries of China, while thin red solid lines indicate known mapped faults. The red hexagonal stars mark the m_L 7.4 Haicheng earthquake on February 4, 1975, and the m_L 7.9 Tangshan earthquake on July 28, 1976. The red solid circle marks the m_L 7.4 Bohai earthquake on July 18, 1969, which generated a prominent aftershock cluster. Contours delineate the spatial extent of clusters that encompass the three mainshocks at density thresholds $f_{\text{th}} = 0.05$ (light green), $f_{\text{th}} = 0.15$ (medium green), and $f_{\text{th}} = 0.25$ (dark green). (Right top) The magnitude-time plot of 60 earthquakes with intensity magnitude $m_i \geq 4.8$ that occurred from 200 to 1900 and (Right bottom) the magnitude-time plot of approximately 3,900 earthquakes with $m_L \geq 3.0$ that occurred from 1900 to 2024 in the vicinity of Bohai Bay, China.

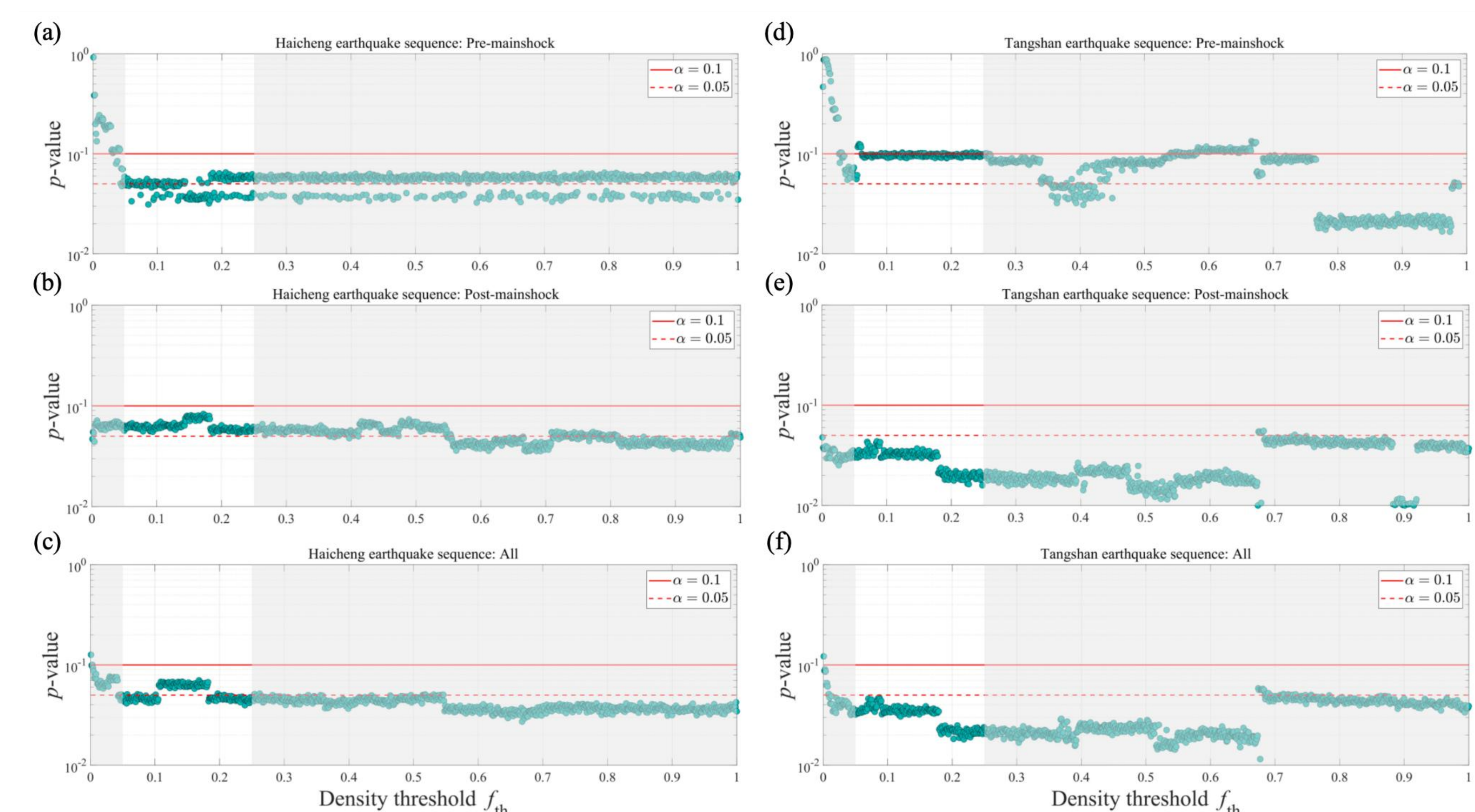


Fig. 3 For the sequences of (a) pre-mainshock, (b) post-mainshock, and (c) combined pre-mainshock and post-mainshock sequences of the m_L 7.4 Haicheng earthquake on February 4, 1975, the p -value of the Haicheng earthquake as a potential dragon-king earthquake is plotted as function of the density threshold f_{th} . (d), (e), and (f) are same as (a) to (c) for the m_L 7.9 Tangshan earthquake on July 28, 1976. The red solid and dashed lines represent the confidence levels ($\alpha = 0.1$ and $\alpha = 0.05$, respectively) commonly used in previous studies. In the white shaded area (for f_{th} approximately between 0.05 and 0.25), we checked that the largest clusters of seismicity are stable and robust.

Key Points:

- Using adaptive kernel density estimation, we propose an objective clustering method that partitions the seismic field into distinct regions.
- We develop a framework to test the dragon-king earthquake hypothesis, using spatial clustering and a high-power outlier detection technique.
- Dragon-king signatures in the Haicheng and Tangshan mainshocks suggest some large and great quakes mature in ways that boost predictability.

Related articles

Li, J., & Sornette, D. (2025). Are Haicheng and Tangshan Earthquakes Dragon-Kings?. (under review) or arXiv preprint arXiv:2504.21310.

Li, J., Sornette, D., Wu, Z., Li, H. (2025). New horizon in the statistical physics of earthquakes: Dragon-king theory and dragon-king earthquakes. Reviews of Geophysics and Planetary Physics, 56(3): 225-242 (in Chinese). doi:10.19975/j.dqyxx.2024-033.

Main references

- Sornette D. 2009. Dragon-kings, black swans and the prediction of crises. Int J Terraspace Sci Eng, 2: 1-18. <https://ssrn.com/abstract=1596032> or <http://dx.doi.org/10.2139/ssrn.1596032>.
- Sornette D. 1999. Complexity, catastrophe and physics. Phys World, 12: 57. 10.1088/2058-7058/12/12/19.
- de S Cavalcante H L D, Oriá M, Sornette D, et al. 2013. Predictability and control of extreme events in complex systems. Phys Rev Lett, 111: 198701. <https://doi.org/10.1103/PhysRevLett.111.198701>.
- Sornette D, Ouillon G. 2012. Dragon-kings: Mechanisms, statistical methods and empirical evidence. Eur Phys J Spec Top, 205: 1-26. <https://doi.org/10.1140/epjst/e2012-01559-5>.
- Sornette D, Wei R. 2025. Multiple outlier detection in samples with exponential and Pareto tails. J Appl Stat., 1-33. <https://doi.org/10.1080/02664763.2025.2511934>.